

Semester – VII

Subject: MACHINE LEARNING

Assignment – 4

Marks: 64

 SECTION – A [$2 \times 5 = 10$ Marks]

(2 Marks Questions – Answer in brief)

Q1. CO4: Define an artificial neuron.

An artificial neuron is a basic unit of a neural network that receives inputs, multiplies them with weights, adds a bias, and passes the result through an activation function to produce an output.

Q2. CO4: What is the perceptron learning rule?

The perceptron learning rule updates the weights of a perceptron based on the error between the actual output and desired output.

It modifies weights so that the model learns to correctly classify training examples.

Q3. CO4: List any two advantages of using multilayer neural networks.

- They can learn complex and non-linear relationships.
 - Can solve problems that single-layer perceptrons cannot, such as XOR classification.
-

Q4. CO5: What is the purpose of the activation function in neural networks?

The activation function introduces non-linearity into the model, allowing the network to learn complex patterns instead of only linear relationships.

Q5. CO4: Define gradient descent.

Gradient descent is an optimization algorithm used to minimize the error (loss) of a model by updating weights in the direction of steepest decrease of the loss function.

 SECTION – B [$3 \times 7 = 21$ Marks]

(7 Marks – Short Essays)

Q1. CO4: Explain the structure and working principle of a perceptron with a neat diagram.

- A perceptron is the simplest form of an artificial neural network containing:
 - **Inputs** (features)
 - **Weights** (importance of each input)
 - **Bias**
 - **Activation function**
 - **Output**
 - **Working steps:**
 1. Each input value is multiplied by its weight.
 2. Weighted inputs are added with bias.
 3. The sum is passed through an activation function (usually sign or step function).
 4. Final output is either 0/1 or -1/+1 based on threshold.
-

Q2. CO4: Discuss the major issues and problems involved in neural network representation.

- Requires large amount of training data
 - Training may be slow and computationally expensive
 - Difficult to interpret and understand internal working (black box problem)
 - Risk of overfitting if network is too large
 - Needs tuning of many parameters (learning rate, number of layers, epochs etc.)
 - Sensitive to noisy or incomplete data
-

Q3. CO4: Describe the backpropagation algorithm and its applications in training multilayer networks.

- Backpropagation is a supervised learning algorithm used in multilayer neural networks.
 - It works in two phases:
 1. **Forward Pass:** Inputs pass through the network to generate output.
 2. **Backward Pass:** Error is calculated and sent backward to update weights using gradient descent.
 - **Applications:**
 - Handwriting recognition
 - Image classification
 - Speech recognition
 - Medical diagnosis systems
-

✓ SECTION – C [$3 \times 11 = 33$ Marks]

(11 Marks – Long Essays)

Q1. CO4: Explain in detail the backpropagation algorithm, including mathematical formulation, learning rate, and convergence issues.

- **Backpropagation** is used for training multilayer neural networks by reducing error through weight updates.
 - **Steps:**
 1. Initialize weights randomly
 2. Perform forward propagation to compute output
 3. Calculate error (difference between actual and predicted output)
 4. Compute gradients of error w.r.t weights
 5. Update weights using gradient descent
 - **Weight Update Formula:**

$$\text{New weight} = \text{Old weight} - (\text{Learning rate} \times \text{Gradient})$$
 - **Learning rate:**
 Controls how fast weights change.
 - Too high → unstable learning
 - Too low → very slow learning
 - **Convergence issues:**
 - May get stuck in local minima
 - Needs proper tuning of learning rate
 - Requires many iterations (epochs)
-

Q2. CO5: Describe the gradient descent method used for training neural networks and differentiate between batch and stochastic gradient descent.

- **Gradient Descent:** Optimization technique used to minimize loss/error by adjusting weights.
 - **Process:**
 - Compute gradient (slope) of loss function
 - Move weights in opposite direction of gradient
 - **Types:**
 1. **Batch Gradient Descent**
 - Uses **all training samples** to calculate error and update weights
 - More stable but slow
 2. **Stochastic Gradient Descent (SGD)**
 - Updates weights **after each training sample**
 - Faster but may fluctuate in learning
-

Q3. CO4: Discuss the architecture, working, and applications of multilayer feedforward neural networks with an example.

- **Architecture:**
 - Input layer
 - One or more hidden layers
 - Output layer
 - Connections move only forward → no feedback loop
- **Working:**
 1. Inputs pass to hidden layers
 2. Each layer applies weighted sum + activation function
 3. Output layer produces final result
 4. Learning is done using backpropagation
- **Applications:**
 - Face recognition

- Self-driving cars
 - Email spam detection
 - Medical prediction systems
- **Example:**
Neural network predicting whether a tumor is cancerous based on medical features.